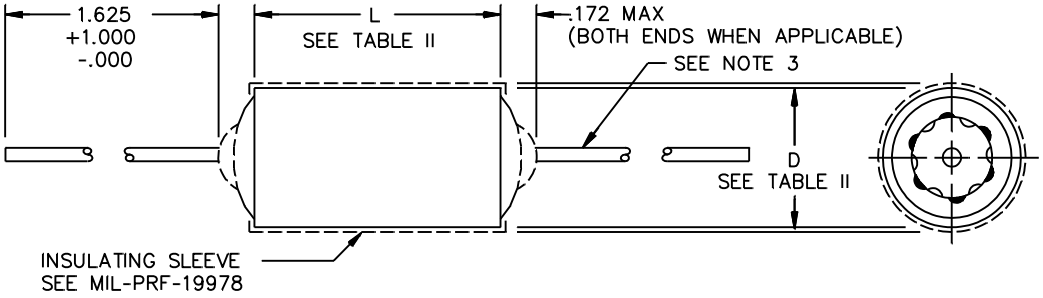
INCH-POUND
MIL-PRF-19978/9D
20 July 2016
SUPERSEDING
MIL-PRF-19978/9C
w/AMENDMENT 1
11 March 2004

PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TUBULAR (INSULATED) (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CQR09

This specification sheet is approved for use by all Departments and Agencies of Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19978.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .312 | 7.92 |
| .004 | 0.10 | .400 | 10.16 |
| .025 | 0.64 | .562 | 14.27 |
| .032 | 0.81 | 1.000 | 25.40 |
| .172 | 4.37 | 1.562 | 39.67 |
| .235 | 5.97 | 1.625 | 41.38 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Leads shall be of solder coated solid wire, .025 inch (No. 22 AWG) for cases .235 inch and .312 inch in diameter; and .032 inch (No. 20 AWG) for cases .400 inch diameter and above. Tolerance on all lead wire diameters shall be +.004 inch, -.001 inch.
4. Capacitors with dimension L of 1.562 inch or D of .562 inch and larger, are not intended to be supported by their leads. These capacitors shall be supported with a supplementary means of mounting, such as a wrap-around band. The supporting device will not be supplied with the capacitor.
5. Lead length may be a minimum of 1-inch long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Style CQR09 capacitors.



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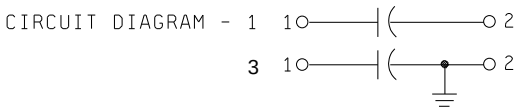
REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table II.

Case: Nonmagnetic (end seal may be of magnetic material).

Terminals: Axial-wire lead (see figure 1).




Rated voltage: See table II.

Rated temperature: -65°C to +125°C.

Capacitance (Cap.) (nom): See table II.

Capacitance tolerance: See table II.

Dissipation factor (DF) (max): 1.0 percent.

Failure rate level: In accordance with MIL-PRF-19978.

Burn-in: In accordance with MIL-PRF-19978.

Radiographic inspection: In accordance with MIL-PRF-19978.

Seal: Method 112 of MIL-STD-202, test condition letter A.

Dielectric withstanding voltage (DWV):

Sleeving: In accordance with MIL-PRF-19978.

Barometric pressure: In accordance with MIL-PRF-19978.

Test points:

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

Insulation resistance (IR):

Sleeving: In accordance with MIL-PRF-19978.

Terminal to terminal: See table I.

Terminal to case: Greater than 10,000 megohms.



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TABLE I. Terminal-to-terminal insulation resistance.

| Capacitance rating | Minimum insulation resistance |
|---|---|
| Characteristic K | |
| | At +25°C |
| 0 to 0.6 microfarad | 25,000 megohms |
| Greater than 0.6 microfarad | 15,000 megohm-microfarads 1/ |
| | At +125°C |
| 0 to 0.08 microfarad | 250 megohms |
| Greater than 0.08 microfarad | 20 megohm-microfarads 1/ |

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray (corrosion): In accordance with MIL-PRF-19978.

Immersion:

DWV:

Sleeving: In accordance with MIL-PRF-19978.

IR:

Sleeving: In accordance with MIL-PRF-19978.

Solderability: In accordance with MIL-PRF-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter D.

Moisture resistance:

DWV:

Sleeving: In accordance with MIL-PRF-19978.

IR:

Sleeving: In accordance with MIL-PRF-19978.

Stability at low and high temperatures:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Capacitance change (max): 10 percent.



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High temperature:

Test temperature: +125°C +0°C, -5°C.

Capacitance change (max): +10 percent.

Life:

Capacitance change (max): ±5 percent of initial measured value.

Resistance to soldering heat: In accordance with MIL-PRF-19978.

TABLE II. Style CQR09 capacitors.

| Part or identifying number (PIN) 1/ | DC voltage rating | Capacitance rating | Capacitance Tolerance | Failure rate level | Case dimensions 2/ | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Circuit 1 | | Circuit 3 | |
| | | | | | L ±.031 | D +.015 -.005 | L ±.031 | D +.015 -.005 |
| | Volts | µF | | | Inches | Inches | Inches | Inches |
| CQR09A-KC392-3- | 200 | .0039 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KC472-3- | 200 | .0047 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KC562-3- | 200 | .0056 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KC682-3- | 200 | .0068 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KC183-3- | 200 | .018 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KC223-3- | 200 | .022 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KC273-3- | 200 | .027 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KC333-3- | 200 | .033 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KC393-3- | 200 | .039 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KC473-3- | 200 | .047 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KC563-3- | 200 | .056 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KC683-3- | 200 | .068 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KC823-3- | 200 | .082 | G, J, K | M, P, R, S | 1.125 | .400 | 1.062 | .400 |
| CQR09A-KC104-3- | 200 | .10 | G, J, K | M, P, R, S | 1.125 | .400 | 1.062 | .400 |
| CQR09A-KC124-3- | 200 | .12 | G, J, K | M, P, R, S | 1.375 | .400 | 1.312 | .400 |
| CQR09A-KC154-3- | 200 | .15 | G, J, K | M, P, R, S | 1.375 | .400 | 1.312 | .400 |
| CQR09A-KC184-3- | 200 | .18 | G, J, K | M, P, R, S | 1.125 | .562 | 1.062 | .562 |
| CQR09A-KC224-3- | 200 | .22 | G, J, K | M, P, R, S | 1.125 | .562 | 1.062 | .562 |
| CQR09A-KC274-3- | 200 | .27 | G, J, K | M, P, R, S | 1.375 | .562 | 1.312 | .562 |
| CQR09A-KC334-3- | 200 | .33 | G, J, K | M, P, R, S | 1.375 | .562 | 1.312 | .562 |
| CQR09A-KC394-3- | 200 | .39 | G, J, K | M, P, R, S | 1.625 | .562 | 1.562 | .562 |
| CQR09A-KC474-3- | 200 | .47 | G, J, K | M, P, R, S | 1.625 | .562 | 1.562 | .562 |
| CQR09A-KC564-3- | 200 | .56 | G, J, K | M, P, R, S | 1.625 | .670 | 1.562 | .670 |
| CQR09A-KC684-3- | 200 | .68 | G, J, K | M, P, R, S | 1.625 | .670 | 1.562 | .670 |
| CQR09A-KC824-3- | 200 | .82 | G, J, K | M, P, R, S | 2.125 | .750 | 2.062 | .750 |
| CQR09A-KC105-3- | 200 | 1.000 | G, J, K | M, P, R, S | 2.125 | .750 | 2.062 | .750 |
| CQR09A-KE272-3- | 400 | .0027 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KE332-3- | 400 | .0033 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KE123-3- | 400 | .012 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KE153-3- | 400 | .015 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KE273-3- | 400 | .027 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KE333-3- | 400 | .033 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KE393-3- | 400 | .039 | G, J, K | M, P, R, S | 1.125 | .400 | 1.062 | .400 |
| CQR09A-KE473-3- | 400 | .047 | G, J, K | M, P, R, S | 1.125 | .400 | 1.062 | .400 |
| CQR09A-KE563-3- | 400 | .056 | G, J, K | M, P, R, S | 1.375 | .400 | 1.312 | .400 |
| CQR09A-KE683-3- | 400 | .068 | G, J, K | M, P, R, S | 1.375 | .400 | 1.312 | .400 |

See footnotes at end of table.



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TABLE II. Style CQR09 capacitors - Continued.

| Part or identifying number (PIN) 1/ | DC voltage rating | Capacitance rating | Capacitance Tolerance | Failure rate level | Case dimensions 2/ | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Circuit 1 | | Circuit 3 | |
| | | | | | L ±.031 | D +.015 -.005 | L ±.031 | D +.015 -.005 |
| | Volts | µF | | | Inches | Inches | Inches | Inches |
| CQR09A-KE823-3- | 400 | .082 | G, J, K | M, P, R, S | 1.125 | .562 | 1.062 | .562 |
| CQR09A-KE104-3- | 400 | .10 | G, J, K | M, P, R, S | 1.125 | .562 | 1.062 | .562 |
| CQR09A-KE124-3- | 400 | .12 | G, J, K | M, P, R, S | 1.375 | .562 | 1.312 | .562 |
| CQR09A-KE154-3- | 400 | .15 | G, J, K | M, P, R, S | 1.375 | .562 | 1.312 | .562 |
| CQR09A-KE184-3- | 400 | .18 | G, J, K | M, P, R, S | 1.625 | .562 | 1.562 | .562 |
| CQR09A-KE224-3- | 400 | .22 | G, J, K | M, P, R, S | 1.625 | .562 | 1.562 | .562 |
| CQR09A-KE274-3- | 400 | .27 | G, J, K | M, P, R, S | 1.625 | .670 | 1.562 | .670 |
| CQR09A-KE334-3- | 400 | .33 | G, J, K | M, P, R, S | 1.625 | .670 | 1.562 | .670 |
| CQR09A-KE394-3- | 400 | .39 | G, J, K | M, P, R, S | 2.125 | .750 | 2.062 | .750 |
| CQR09A-KE474-3- | 400 | .47 | G, J, K | M, P, R, S | 2.125 | .750 | 2.062 | .750 |
| CQR09A-KF102-3- | 600 | .0010 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KF122-3- | 600 | .0012 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KF152-3- | 600 | .0015 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KF182-3- | 600 | .0018 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KF222-3- | 600 | .0022 | G, J, K | M, P, R, S | .750 | .235 | .688 | .235 |
| CQR09A-KF272-3- | 600 | .0027 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KF332-3- | 600 | .0033 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KF392-3- | 600 | .0039 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KF472-3- | 600 | .0047 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KF562-3- | 600 | .0056 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KF682-3- | 600 | .0068 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KF822-3- | 600 | .0082 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KF103-3- | 600 | .010 | G, J, K | M, P, R, S | .875 | .312 | .812 | .312 |
| CQR09A-KF123-3- | 600 | .012 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KF153-3- | 600 | .015 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KF183-3- | 600 | .018 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KF223-3- | 600 | .022 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KF273-3- | 600 | .027 | G, J, K | M, P, R, S | 1.125 | .400 | 1.062 | .400 |
| CQR09A-KF333-3- | 600 | .033 | G, J, K | M, P, R, S | 1.125 | .400 | 1.062 | .400 |
| CQR09A-KF393-3- | 600 | .039 | G, J, K | M, P, R, S | 1.375 | .400 | 1.312 | .400 |
| CQR09A-KF473-3- | 600 | .047 | G, J, K | M, P, R, S | 1.375 | .400 | 1.312 | .400 |
| CQR09A-KF563-3- | 600 | .056 | G, J, K | M, P, R, S | 1.125 | .562 | 1.062 | .562 |
| CQR09A-KF683-3- | 600 | .068 | G, J, K | M, P, R, S | 1.125 | .562 | 1.062 | .562 |
| CQR09A-KF823-3- | 600 | .082 | G, J, K | M, P, R, S | 1.375 | .562 | 1.312 | .562 |
| CQR09A-KF104-3- | 600 | .10 | G, J, K | M, P, R, S | 1.375 | .562 | 1.312 | .562 |
| CQR09A-KF124-3- | 600 | .12 | G, J, K | M, P, R, S | 1.625 | .562 | 1.562 | .562 |
| CQR09A-KF154-3- | 600 | .15 | G, J, K | M, P, R, S | 1.625 | .562 | 1.562 | .562 |
| CQR09A-KF184-3- | 600 | .18 | G, J, K | M, P, R, S | 1.625 | .670 | 1.562 | .670 |
| CQR09A-KF224-3- | 600 | .22 | G, J, K | M, P, R, S | 1.625 | .670 | 1.562 | .670 |
| CQR09A-KF274-3- | 600 | .27 | G, J, K | M, P, R, S | 2.125 | .750 | 2.062 | .750 |
| CQR09A-KF334-3- | 600 | .33 | G, J, K | M, P, R, S | 2.125 | .750 | 2.062 | .750 |
| CQR09A-KF394-3- | 600 | .39 | G, J, K | M, P, R, S | 2.375 | .750 | 2.312 | .750 |
| CQR09A-KF474-3- | 600 | .47 | G, J, K | M, P, R, S | 2.375 | .750 | 2.312 | .750 |
| CQR09A-KG102-3- | 1000 | .0010 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG122-3- | 1000 | .0012 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG152-3- | 1000 | .0015 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG182-3- | 1000 | .0018 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |

See footnotes at end of table.



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TABLE II. Style CQR09 capacitors - Continued.

| Part or identifying number (PIN) 1/ | DC voltage rating | Capacitance rating | Capacitance Tolerance | Failure rate level | Case dimensions 2/ | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Circuit 1 | | Circuit 3 | |
| | | | | | L ±.031 | D +.015 -.005 | L ±.031 | D +.015 -.005 |
| | Volts | µF | | | Inches | Inches | Inches | Inches |
| CQR09A-KG222-3- | 1000 | .0022 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG272-3- | 1000 | .0027 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG332-3- | 1000 | .0033 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG392-3- | 1000 | .0039 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG472-3- | 1000 | .0047 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG562-3- | 1000 | .0056 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG682-3- | 1000 | .0068 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG822-3- | 1000 | .0082 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG103-3- | 1000 | .010 | G, J, K | M, P, R, S | .875 | .400 | .812 | .400 |
| CQR09A-KG123-3- | 1000 | .012 | G, J, K | M, P, R, S | 1.125 | .400 | 1.062 | .400 |
| CQR09A-KG153-3- | 1000 | .015 | G, J, K | M, P, R, S | 1.125 | .400 | 1.062 | .400 |
| CQR09A-KG183-3- | 1000 | .018 | G, J, K | M, P, R, S | 1.375 | .400 | 1.312 | .400 |
| CQR09A-KG223-3- | 1000 | .022 | G, J, K | M, P, R, S | 1.375 | .400 | 1.312 | .400 |
| CQR09A-KG273-3- | 1000 | .027 | G, J, K | M, P, R, S | 1.125 | .562 | 1.062 | .562 |
| CQR09A-KG333-3- | 1000 | .033 | G, J, K | M, P, R, S | 1.125 | .562 | 1.062 | .562 |
| CQR09A-KG393-3- | 1000 | .039 | G, J, K | M, P, R, S | 1.375 | .562 | 1.312 | .562 |
| CQR09A-KG473-3- | 1000 | .047 | G, J, K | M, P, R, S | 1.375 | .562 | 1.312 | .562 |
| CQR09A-KG563-3- | 1000 | .056 | G, J, K | M, P, R, S | 1.625 | .562 | 1.562 | .562 |
| CQR09A-KG683-3- | 1000 | .068 | G, J, K | M, P, R, S | 1.625 | .562 | 1.562 | .562 |
| CQR09A-KG823-3- | 1000 | .082 | G, J, K | M, P, R, S | 1.625 | .670 | 1.562 | .670 |
| CQR09A-KG104-3- | 1000 | .10 | G, J, K | M, P, R, S | 1.625 | .670 | 1.562 | .670 |
| CQR09A-KG124-3- | 1000 | .12 | G, J, K | M, P, R, S | 1.875 | .670 | 1.812 | .670 |
| CQR09A-KG154-3- | 1000 | .15 | G, J, K | M, P, R, S | 1.875 | .670 | 1.812 | .670 |
| CQR09A-KG184-3- | 1000 | .18 | G, J, K | M, P, R, S | 2.125 | .750 | 2.062 | .750 |
| CQR09A-KG224-3- | 1000 | .22 | G, J, K | M, P, R, S | 2.125 | .750 | 2.062 | .750 |
| CQR09A-KG274-3- | 1000 | .27 | G, J, K | M, P, R, S | 2.125 | 1.000 | 2.062 | 1.000 |
| CQR09A-KG334-3- | 1000 | .33 | G, J, K | M, P, R, S | 2.125 | 1.000 | 2.062 | 1.000 |
| CQR09A-KG394-3- | 1000 | .39 | G, J, K | M, P, R, S | 2.375 | 1.000 | 2.312 | 1.000 |
| CQR09A-KG474-3- | 1000 | .47 | G, J, K | M, P, R, S | 2.375 | 1.000 | 2.312 | 1.000 |

1/ Complete part number shall include additional symbols to indicate circuit, capacitance tolerance and failure rate level as applicable.

2/ Dimensions are for basic case; sleeving shall extend .016 inch minimum and .062 inch maximum, beyond each end of the capacitor body; however, if a shrink-fitted insulation is used for the sleeving, it shall lap over the ends of the capacitor body. Add .047 inch maximum to the nominal for capacitor diameter. See table III for metric equivalents.

TABLE III. Millimeter equivalent of decimal inches.

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .005 | 0.13 | .562 | 14.27 | 1.312 | 33.32 |
| .015 | 0.38 | .670 | 17.02 | 1.375 | 34.93 |
| .016 | 0.41 | .688 | 17.48 | 1.562 | 39.67 |
| .031 | 0.79 | .750 | 19.05 | 1.625 | 41.28 |
| .047 | 1.19 | .812 | 20.62 | 2.062 | 52.37 |
| .062 | 1.57 | .875 | 22.23 | 2.125 | 53.98 |
| .235 | 5.97 | 1.000 | 25.40 | 2.312 | 58.72 |
| .312 | 7.92 | 1.062 | 26.97 | 2.375 | 60.33 |
| .400 | 10.16 | 1.125 | 28.58 | | |



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Referenced documents: In addition to MIL-PRF-19978, this document references the following:

MIL-STD-202-112
MIL-STD-202-204
MIL-STD-202-211

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2016-022)

Review activities:
Navy - MC
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.



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